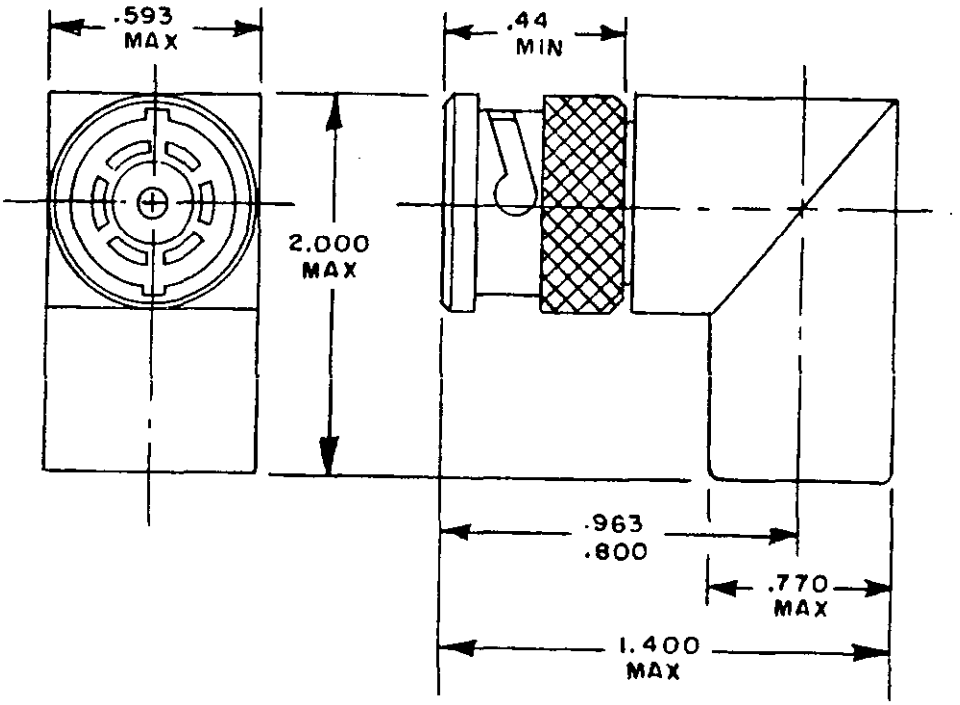
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY, (SERIES TRB, (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-49142.

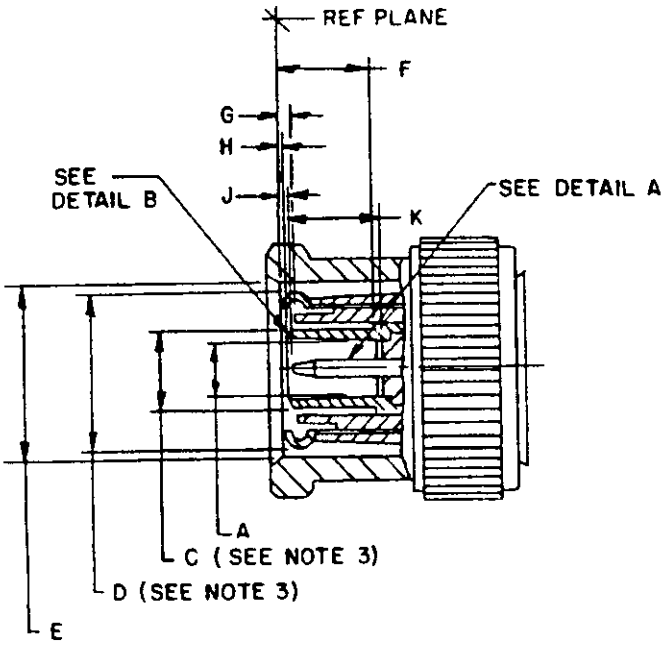


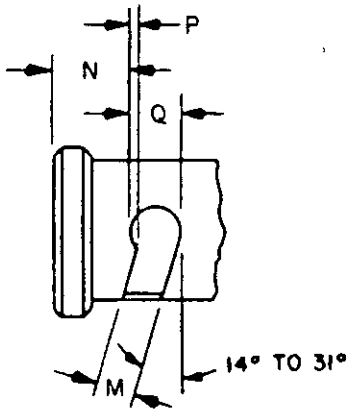
INCHES	MM
.44	11.18
.510	12.95
.593	15.06
.770	19.56
.800	20.32
.963	24.46
1.400	35.56
2.000	50.80

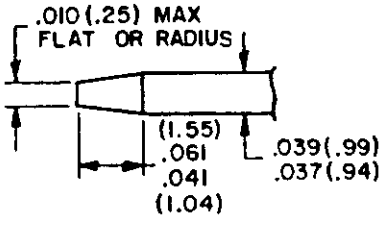
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. 2.000 maximum defines the maximum length of the connector when assembled to the appropriate cable.
4. Wrench flats are to accommodate standard wrench openings per FED-STD-H28.
5. All undimensioned pictorial representations are for reference purposes only.

FIGURE 1. General configuration.

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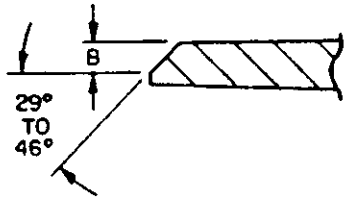




CENTER CONDUCTOR
DETAIL A

Dimensions

Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.123	---	3.12	---
B	.005	.010	.13	.25
C	.172	.178	4.37	4.52
D	.337	.347	8.56	8.81
E	.385	.390	9.78	9.91
F	.213	.240	5.41	6.10
G	.001	---	.03	---
H	.007	.031	.18	.79
J	.018	.046	.46	1.17
K	.210	---	5.33	---
M	.091	.097	2.31	2.46
N	.168	.172	4.27	4.37
P	.018	.022	.46	.56
Q	.124	---	3.15	---



INTERMEDIATE CONDUCTOR
DETAIL B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Flared to meet mating characteristic test.
4. Metric equivalents are in parentheses or tabulated.

FIGURE 2. Mating dimensions for pin contacts.

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ENGINEERING DATA:

Nominal impedance: Non constant.

Frequency range: 0 to 500 MHz, minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature rating: -55° to +165°C.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Force to engage and disengage:
Longitudinal force - 4 pounds, maximum.
Torque - 2.5 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See figure 2 for dimensions.
Intermediate contact:
Test ring ID - .169 inch maximum, 16 microinch finish.
Insertion force - 3 pounds maximum, when inserted a minimum of .093.
Contacts with slotted members: Shall contact a .173 inch minimum diameter ring within .031 inch of their tip ends.
Outer contact:
Test ring ID - .319 inch maximum, 16 microinch finish.
Insertion force - 5 pounds maximum, when inserted a minimum of .093 inch.
Contacts with slotted members: Shall contact a .324 inch diameter ring within .031 inch of their tip ends.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: To be added.

Insulation resistance: 5000 megohms.

Outer conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage: At sea level, 1200 V rms, between center conductor and intermediate conductor, 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.2	0.3

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TABLE I. Group Qualification.

Group	Submission and qualification of any of the following dash numbers	Qualifies the following dash numbers
I	-0001 -0002 -0005	-0001 -0002 -0005
II	-0003 -0004	-0003 -0004
III	-0006	-0006

Corona level:
Altitude - 70,000 feet.
Voltage - 250 V rms, minimum.

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension): 40 pounds, minimum.

Coupling mechanism retention force: 100 pounds, minimum.

Risetime degradation: To be added.

Connector durability: 500 cycles minimum, at 12 cycles per minute maximum.

Part number: M49142/07- (dash number from table II)

TABLE II. Dash numbers, cross reference to cable.

Dash number 1/	Cross reference number
0001	M17/134-00001
0002	M17/134-00002
0003	D3-7619-5/336
0004	D3-7619-5/338
0005	M17/116-RG307
0006 2/	M17/176-00002

1/ Not for Naval Air Systems Command (AS) use.
2/ Not for Air Force use.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3359-1)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49142/7	Connectors, Plug, Electrical, Triaxial, Radio Frequency(Series TRB, (Cabled), Pin Contact, Right Angle, Class 2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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